

SSS 1-3

12-WEEK TEACHER LESSON PLAN

FRAME -> DESIGN -> INSTRUMENT -> MODEL -> VALIDATE -> OPTIMIZE

Structure	Weeks 1-10: one GrowMeal garden per week. Week 11: synthesis, remediation and portfolio. Week 12: celebration, demonstration and level completion.
Integrated resources	Vocabulary -> Flashcards -> Teacher Demonstration -> Practical Activity -> Journal -> Garden Mission -> Ingredient Lab/Farm-to-Food/Food Smart -> Assessment -> Passport.
Teacher preparation	Inspect the garden/site, choose exact cards/activities, prepare materials and DIY substitutes, review safety, assign learner groups and pre-check instruments.
Evidence rule	Every week should leave visible evidence: oral/drawn evidence at Nursery; records and measurements in Primary; increasingly traceable data, analysis and decisions at JSS/SSS.
Safety	School safeguarding, food-safety and tool-use rules take priority. No tasting is required. Teachers control knives, heat, chemicals, powered equipment and other higher-risk tools.

WEEK 01 - Advanced Production Garden

Learning objective	Design and evaluate a research/engineering investigation in the Advanced Production Garden, including measurement quality, uncertainty and an optimization decision.
Vocabulary	15-20 research, engineering, instrumentation, modeling and systems terms used in technical context.
Resources	Flashcards for Advanced Production Garden; TD-401 to TD-410; PA-401 to PA-410; GM-401 to GM-410; AQ-401 to AQ-410.
Materials / DIY	Low-cost reference measurements, paper-based process/system models, locally fabricated sampling devices and manual data logging where digital systems are unavailable.
Suggested lesson flow	Research/design objective (10 min) -> protocol/instrument/QC (20 min) -> data/model work (40 min) -> validation/trade-off analysis (20 min) -> optimization recommendation (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-401–IL-410, FF-401–FF-410, or FSC-401–FSC-410 activity where it fits the week's garden evidence.
Questions to ask	What decision must the evidence support? How was measurement quality checked? Which assumption dominates? What trade-off limits optimization? How will the recommendation be validated?
Assessment	Use observation plus AQ-401–AQ-410; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 02 - Crop Trial Garden

Learning objective	Design and evaluate a research/engineering investigation in the Crop Trial Garden, including measurement quality, uncertainty and an optimization decision.
Vocabulary	15-20 research, engineering, instrumentation, modeling and systems terms used in technical context.
Resources	Flashcards for Crop Trial Garden; TD-411 to TD-420; PA-411 to PA-420; GM-411 to GM-420; AQ-411 to AQ-420.
Materials / DIY	Low-cost reference measurements, paper-based process/system models, locally fabricated sampling devices and manual data logging where digital systems are unavailable.
Suggested lesson flow	Research/design objective (10 min) -> protocol/instrument/QC (20 min) -> data/model work (40 min) -> validation/trade-off analysis (20 min) -> optimization recommendation (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-411–IL-420, FF-411–FF-420, or FSC-411–FSC-420 activity where it fits the week's garden evidence.
Questions to ask	What decision must the evidence support? How was measurement quality checked? Which assumption dominates? What trade-off limits optimization? How will the recommendation be validated?
Assessment	Use observation plus AQ-411–AQ-420; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 03 - Controlled Experiment Garden

Learning objective	Design and evaluate a research/engineering investigation in the Controlled Experiment Garden, including measurement quality, uncertainty and an optimization decision.
Vocabulary	15-20 research, engineering, instrumentation, modeling and systems terms used in technical context.
Resources	Flashcards for Controlled Experiment Garden; TD-421 to TD-430; PA-421 to PA-430; GM-421 to GM-430; AQ-421 to AQ-430.
Materials / DIY	Low-cost reference measurements, paper-based process/system models, locally fabricated sampling devices and manual data logging where digital systems are unavailable.
Suggested lesson flow	Research/design objective (10 min) -> protocol/instrument/QC (20 min) -> data/model work (40 min) -> validation/trade-off analysis (20 min) -> optimization recommendation (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-421–IL-430, FF-421–FF-430, or FSC-421–FSC-430 activity where it fits the week's garden evidence.
Questions to ask	What decision must the evidence support? How was measurement quality checked? Which assumption dominates? What trade-off limits optimization? How will the recommendation be validated?
Assessment	Use observation plus AQ-421–AQ-430; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 04 - Irrigation Engineering Garden

Learning objective	Design and evaluate a research/engineering investigation in the Irrigation Engineering Garden, including measurement quality, uncertainty and an optimization decision.
Vocabulary	15-20 research, engineering, instrumentation, modeling and systems terms used in technical context.
Resources	Flashcards for Irrigation Engineering Garden; TD-431 to TD-440; PA-431 to PA-440; GM-431 to GM-440; AQ-431 to AQ-440.
Materials / DIY	Low-cost reference measurements, paper-based process/system models, locally fabricated sampling devices and manual data logging where digital systems are unavailable.
Suggested lesson flow	Research/design objective (10 min) -> protocol/instrument/QC (20 min) -> data/model work (40 min) -> validation/trade-off analysis (20 min) -> optimization recommendation (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-431–IL-440, FF-431–FF-440, or FSC-431–FSC-440 activity where it fits the week's garden evidence.
Questions to ask	What decision must the evidence support? How was measurement quality checked? Which assumption dominates? What trade-off limits optimization? How will the recommendation be validated?
Assessment	Use observation plus AQ-431–AQ-440; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 05 - Water Productivity Garden

Learning objective	Design and evaluate a research/engineering investigation in the Water Productivity Garden, including measurement quality, uncertainty and an optimization decision.
Vocabulary	15-20 research, engineering, instrumentation, modeling and systems terms used in technical context.
Resources	Flashcards for Water Productivity Garden; TD-441 to TD-450; PA-441 to PA-450; GM-441 to GM-450; AQ-441 to AQ-450.
Materials / DIY	Low-cost reference measurements, paper-based process/system models, locally fabricated sampling devices and manual data logging where digital systems are unavailable.
Suggested lesson flow	Research/design objective (10 min) -> protocol/instrument/QC (20 min) -> data/model work (40 min) -> validation/trade-off analysis (20 min) -> optimization recommendation (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-441–IL-450, FF-441–FF-450, or FSC-441–FSC-450 activity where it fits the week's garden evidence.
Questions to ask	What decision must the evidence support? How was measurement quality checked? Which assumption dominates? What trade-off limits optimization? How will the recommendation be validated?
Assessment	Use observation plus AQ-441–AQ-450; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 06 - Advanced Soilless Garden

Learning objective	Design and evaluate a research/engineering investigation in the Advanced Soilless Garden, including measurement quality, uncertainty and an optimization decision.
Vocabulary	15-20 research, engineering, instrumentation, modeling and systems terms used in technical context.
Resources	Flashcards for Advanced Soilless Garden; TD-451 to TD-460; PA-451 to PA-460; GM-451 to GM-460; AQ-451 to AQ-460.
Materials / DIY	Low-cost reference measurements, paper-based process/system models, locally fabricated sampling devices and manual data logging where digital systems are unavailable.
Suggested lesson flow	Research/design objective (10 min) -> protocol/instrument/QC (20 min) -> data/model work (40 min) -> validation/trade-off analysis (20 min) -> optimization recommendation (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-451–IL-460, FF-451–FF-460, or FSC-451–FSC-460 activity where it fits the week's garden evidence.
Questions to ask	What decision must the evidence support? How was measurement quality checked? Which assumption dominates? What trade-off limits optimization? How will the recommendation be validated?
Assessment	Use observation plus AQ-451–AQ-460; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 07 - Sensor & Monitoring Garden

Learning objective	Design and evaluate a research/engineering investigation in the Sensor & Monitoring Garden, including measurement quality, uncertainty and an optimization decision.
Vocabulary	15-20 research, engineering, instrumentation, modeling and systems terms used in technical context.
Resources	Flashcards for Sensor & Monitoring Garden; TD-461 to TD-470; PA-461 to PA-470; GM-461 to GM-470; AQ-461 to AQ-470.
Materials / DIY	Low-cost reference measurements, paper-based process/system models, locally fabricated sampling devices and manual data logging where digital systems are unavailable.
Suggested lesson flow	Research/design objective (10 min) -> protocol/instrument/QC (20 min) -> data/model work (40 min) -> validation/trade-off analysis (20 min) -> optimization recommendation (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-461–IL-470, FF-461–FF-470, or FSC-461–FSC-470 activity where it fits the week's garden evidence.
Questions to ask	What decision must the evidence support? How was measurement quality checked? Which assumption dominates? What trade-off limits optimization? How will the recommendation be validated?
Assessment	Use observation plus AQ-461–AQ-470; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 08 - Smart Irrigation Demonstrator

Learning objective	Design and evaluate a research/engineering investigation in the Smart Irrigation Demonstrator, including measurement quality, uncertainty and an optimization decision.
Vocabulary	15-20 research, engineering, instrumentation, modeling and systems terms used in technical context.
Resources	Flashcards for Smart Irrigation Demonstrator; TD-471 to TD-480; PA-471 to PA-480; GM-471 to GM-480; AQ-471 to AQ-480.
Materials / DIY	Low-cost reference measurements, paper-based process/system models, locally fabricated sampling devices and manual data logging where digital systems are unavailable.
Suggested lesson flow	Research/design objective (10 min) -> protocol/instrument/QC (20 min) -> data/model work (40 min) -> validation/trade-off analysis (20 min) -> optimization recommendation (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-471–IL-480, FF-471–FF-480, or FSC-471–FSC-480 activity where it fits the week's garden evidence.
Questions to ask	What decision must the evidence support? How was measurement quality checked? Which assumption dominates? What trade-off limits optimization? How will the recommendation be validated?
Assessment	Use observation plus AQ-471–AQ-480; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 09 - Research & Design Garden

Learning objective	Design and evaluate a research/engineering investigation in the Research & Design Garden, including measurement quality, uncertainty and an optimization decision.
Vocabulary	15-20 research, engineering, instrumentation, modeling and systems terms used in technical context.
Resources	Flashcards for Research & Design Garden; TD-481 to TD-490; PA-481 to PA-490; GM-481 to GM-490; AQ-481 to AQ-490.
Materials / DIY	Low-cost reference measurements, paper-based process/system models, locally fabricated sampling devices and manual data logging where digital systems are unavailable.
Suggested lesson flow	Research/design objective (10 min) -> protocol/instrument/QC (20 min) -> data/model work (40 min) -> validation/trade-off analysis (20 min) -> optimization recommendation (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-481–IL-490, FF-481–FF-490, or FSC-481–FSC-490 activity where it fits the week's garden evidence.
Questions to ask	What decision must the evidence support? How was measurement quality checked? Which assumption dominates? What trade-off limits optimization? How will the recommendation be validated?
Assessment	Use observation plus AQ-481–AQ-490; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 10 - Integrated Food Systems Laboratory

Learning objective	Design and evaluate a research/engineering investigation in the Integrated Food Systems Laboratory, including measurement quality, uncertainty and an optimization decision.
Vocabulary	15-20 research, engineering, instrumentation, modeling and systems terms used in technical context.
Resources	Flashcards for Integrated Food Systems Laboratory; TD-491 to TD-500; PA-491 to PA-500; GM-491 to GM-500; AQ-491 to AQ-500.
Materials / DIY	Low-cost reference measurements, paper-based process/system models, locally fabricated sampling devices and manual data logging where digital systems are unavailable.
Suggested lesson flow	Research/design objective (10 min) -> protocol/instrument/QC (20 min) -> data/model work (40 min) -> validation/trade-off analysis (20 min) -> optimization recommendation (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-491–IL-500, FF-491–FF-500, or FSC-491–FSC-500 activity where it fits the week's garden evidence.
Questions to ask	What decision must the evidence support? How was measurement quality checked? Which assumption dominates? What trade-off limits optimization? How will the recommendation be validated?
Assessment	Use observation plus AQ-491–AQ-500; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 11 - Synthesis & Portfolio

Learning objective	Synthesize evidence from the ten gardens, revisit weak competencies, complete unfinished missions, and prepare a portfolio of the learner's strongest evidence.
Vocabulary	15-20 research, engineering, instrumentation, modeling and systems terms used in technical context.
Resources	Student Journal; Food Smart Passport; selected Practical Cards, Garden Missions, Ingredient Lab, Farm-to-Food and Food Smart Challenge evidence from Weeks 1-10.
Materials / DIY	Low-cost reference measurements, paper-based process/system models, locally fabricated sampling devices and manual data logging where digital systems are unavailable.
Suggested lesson flow	Research/design objective (10 min) -> protocol/instrument/QC (20 min) -> data/model work (40 min) -> validation/trade-off analysis (20 min) -> optimization recommendation (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Learners choose evidence that shows growth in knowledge, practical competence, investigation/data skills and mission-based problem solving.
Questions to ask	What decision must the evidence support? How was measurement quality checked? Which assumption dominates? What trade-off limits optimization? How will the recommendation be validated?
Assessment	Teacher moderation: review journal, passport, assessment results and practical evidence; reteach or reassess competencies not yet secure.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 12 - GrowMeal Celebration & Demonstration

Learning objective	Celebrate learning through a safe GrowMeal showcase in which learners communicate what they grew, observed, measured, investigated and solved.
Vocabulary	15-20 research, engineering, instrumentation, modeling and systems terms used in technical context.
Resources	Food Smart Passport; learner portfolio; garden displays; selected charts/models; certificates or school-approved recognition materials.
Materials / DIY	Low-cost reference measurements, paper-based process/system models, locally fabricated sampling devices and manual data logging where digital systems are unavailable.
Suggested lesson flow	Research/design objective (10 min) -> protocol/instrument/QC (20 min) -> data/model work (40 min) -> validation/trade-off analysis (20 min) -> optimization recommendation (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Garden walk, learner demonstrations, evidence exhibition, Farm-to-Food/Ingredient Lab displays and age-appropriate challenge presentations.
Questions to ask	What decision must the evidence support? How was measurement quality checked? Which assumption dominates? What trade-off limits optimization? How will the recommendation be validated?
Assessment	Final teacher verification, learner reflection, level-completion record and next-level learning goals.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?